INDUSTRIAL INTERNET OF THINGS (IIoT-3)

# Road Accident Pattern Analysis & IIoT Smart Highway Monitoring

## A Comprehensive System for Empirical Risk Analytics, Closed-Loop VMS Actuation & Emergency CAD Dispatch

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| **Course / Domain:** | IIoT-3 • Intelligent Transportation Systems (ITS) | **Date:** | November 2026 |
| **Team Leader:** | Lovish Chauhan (lovishneeraj@gmail.com) | **Team Size:** | 14 Members |
| **GitHub Repo:** | https://github.com/igris-bit/Road-Accident-Pattern-Analysis | **Live App:** | https://road-accident-pattern-analysis.streamlit.app/ |

## 1. Executive Summary

Road traffic collisions account for over 150,000 fatalities annually in India, creating severe socio-economic impact. Traditional transportation management is largely retrospective—relying on post-crash police dossiers and static signboards that fail to avert accidents before impact. This project pioneers a hybrid safety framework that bridges **retrospective empirical data analysis** with **proactive Industrial IoT (IIoT) edge simulation and actuation**.

Calibrated on a ground-truth dataset of **20,000 records across 8 major Indian metropolitan corridors** (*Bangalore, Chandigarh, Chennai, Delhi, Hyderabad, Kolkata, Mumbai, Pune*), the system delivers: (1) A multivariate analytics dashboard featuring an empirical Bayesian 'What-If' accident severity predictor; (2) An industrial IoT edge simulation engine that translates optical, environmental, and speed telemetry into multi-tier hazard alerts; (3) A dynamic overhead Variable Message Sign (VMS) actuator overriding highway speed limits in real time; and (4) An automated Computer-Aided Dispatch (CAD) interconnect routing nearest Level-1 trauma centers (AIIMS, KEM, PGIMER, etc.) within the critical Golden Hour.

## 2. Problem Statement & System Objectives

**Problem Statement:** Arterial highways in India experience violent weather shifts (dense winter fog, monsoon flooding), high vehicle speed differentials, and delayed medical response. Existing traffic monitoring systems lack automated closed-loop feedback between roadside sensors and driver warnings.

**Primary Technical Objectives:**

- **Deep Empirical Analysis:** Identify casualty correlations across time (24h commuting peaks), weather conditions, and road classifications from 20,000 verified records.
- **Bayesian AI Predictor:** Formulate an empirical likelihood model calculating accident severity probabilities (Minor vs. Major vs. Fatal) based on real-time driving inputs.
- **Closed-Loop IIoT Architecture:** Implement the complete 3-tier IIoT paradigm: Perception (Edge Sensors) → Network/Edge Inference → Actuation (VMS Speed Overrides & CAD Ambulance Routing).

- **Cloud CI/CD Deployment:** Standardize production on a single clean entrypoint (streamlit_app.py) running live on Streamlit Community Cloud.

## 3. Dataset Architecture & Preprocessing Pipeline

The project is empirically grounded in indian_roads_dataset.csv (20,000 records × 24 attributes). The dataset encompasses spatial, temporal, environmental, infrastructure, and impact dimensions:

| Feature Dimension | Attributes Included | Engineering / Normalization Applied |
|---|---|---|
| **Spatial Identifiers** | city, state, latitude, longitude | Bounded coordinates strictly to 8 metropolitan regional corridors. |
| **Temporal Markers** | date, time, hour (0-23), day_of_week, is_weekend, is_peak_hour | Engineered peak-hour binary flags (08:00-10:00 & 17:00-20:00). |
| **Road Infrastructure** | road_type (highway/rural/urban), lanes (1-6), traffic_signal | Categorical standardization; cross-verified lane capacity. |
| **Environmental Conditions** | weather (clear/fog/rain), visibility (high/med/low), temperature | Normalized visibility optical scales; temperature in °C. |
| **Impact & Evaluation** | traffic_density, vehicles_involved, casualties, severity, risk_score | Eliminated sparse columns (e.g. festival); risk mapped [0, 1]. |

## 4. Three-Tier Industrial IoT (IIoT) Architecture

Unlike academic dashboards that merely visualize historical CSVs, this system executes an industrial **closed-loop sensing-to-actuation architecture**:

| IIoT Layer | Components & Hardware Roles | Data Flow & Technical Function |
|---|---|---|
| **Tier 1: Perception Layer** (Roadside Edge Sensing) | • Optical Visibility Meters<br>• Roadside Weather Stations<br>• Radar Speed Doppler Units<br>• Inductive Traffic Counters | Collects physical road telemetry: ambient temperature, optical transmission (fog density), precipitation presence, vehicle headway, and average corridor velocity. |
| **Tier 2: Network & Edge Layer** (Edge Computing) | • Edge Gateway Node<br>• Rule-Based Alert Engine<br>• Empirical Bayesian Predictor | Simulates MQTT/REST packet transport. Ingests telemetry every 5s, applies multi-factor hazard escalation, and computes instant severity likelihoods. |
| **Tier 3: Application & Actuation** (Command & Safety) | • Streamlit Community Cloud<br>• Dynamic Highway VMS Actuator<br>• Automated Emergency CAD Interconnect | Renders live operator console. Actuates physical overhead LED Variable Message Signs with speed overrides and auto-dispatches regional trauma units. |

# 5. Detailed Module Implementation

## 5.1 Module 1: Historical Analytics & 'What-If' Severity Predictor (streamlit_app.py)

Module 1 serves as the command analytics center. It incorporates multivariate filtering across State, City, Severity, Road Type, and Weather, alongside a 2×2 visual distribution matrix and Top 10 state rankings. Most importantly, it integrates the **Interactive 'What-If' Scenario Predictor**:

- **Mathematical Calibration:** Estimates conditional severity distributions: *P(Severity | Weather, Visibility, Road, Time, Density)* using Laplace-smoothed empirical Bayes ratios directly from the 20,000 dataset records.
- **Input Matrix:** Corridor City, Road Classification, Weather Condition, Optical Visibility, Traffic Density, Hour of Day (0-23), and Vehicles Involved (1-5).
- **Real-Time Outputs:** (1) Predicted Severity (Minor / Major / Fatal); (2) Severity Probability Distribution (e.g., Minor: 48.2% | Major: 36.5% | Fatal: 15.3%); (3) Normalized Composite Risk Score [0.00, 1.00]; and (4) Automated Contributing Hazard Breakdown.

## 5.2 Module 2: IIoT Edge Telemetry, Smart VMS Actuator & CAD Dispatch (pages/iot_monitor.py)

Module 2 delivers the operational industrial IoT runtime environment:

- **Roadside Telemetry Generation:** Generates realistic sensor packets grounded in the exact distributions of the 8 Indian cities (Bangalore, Chandigarh, Chennai, Delhi, Hyderabad, Kolkata, Mumbai, Pune).
- **Multi-Tier Alert Escalation:** Categorizes hazard severity into SAFE (clear/day), LOW (urban commuter congestion), MEDIUM (rain, daytime low visibility, extreme heat ≥42°C), and HIGH (night fog + low visibility, highway rain + high density, risk ≥0.70).
- **Smart Highway Variable Message Sign (VMS):** Renders an electronic overhead LED matrix billboard styled with retro-illuminated amber/red typography. Automatically dampens corridor speed limits: 80 km/h (Normal) → 50 km/h (Rain) → 35 km/h (Dense Fog / Night Collision Hazard).
- **Computer-Aided Dispatch (CAD) Interconnect:** Automatically assigns the regional Level-1 trauma facility (e.g. AIIMS Trauma Centre for Delhi, KEM for Mumbai, Victoria Hospital for Bangalore, PGIMER for Chandigarh), calculates highway patrol interceptors, and computes real-time ambulance arrival ETAs (5–8 minutes).
- **Geospatial Pydeck Hotspot Mapping:** Renders a 3D interactive Scatterplot layer displaying real accident coordinates alongside the active IoT sensor node, paired with a persistent incident audit log with timestamped CSV export.

# 6. Experimental Findings & Real-World Impact

1. **Adverse Weather Multiplier:** Dense winter fog coupled with low visibility during nighttime hours (21:00-05:00) yields a **+42% increase in fatality likelihood**, representing the single highest hazard factor across all 8 corridors.

2. **Highway vs. Urban Severity Ratio:** Highway collisions exhibit **2.3× higher casualty rates per incident** compared to urban roads due to elevated approach momentum, confirming the necessity of dynamic VMS speed intervention.

3. **Golden Hour Response Optimization:** Automated CAD routing provides an ambulance arrival window of **5–8 minutes**, mitigating emergency notification delays and dramatically improving survival odds for critically injured highway commuters.

# 7. Comprehensive 14-Member Contribution Audit Matrix

Official contribution breakdown audited directly from Git commit hashes, author logs, and remote branches across the repository:

| Member & Branch | Assigned Role & Contributions Delivered | Level |
|---|---|---|
| **Lovish Chauhan**<br>(Team Leader)<br>Branch: Lovish | • Spearheaded system architecture, live deployment on Streamlit Cloud, and repository governance.<br>• Built the 'What-If' ML Accident Severity & Risk Predictor.<br>• Engineered Smart Highway VMS actuator and CAD emergency hospital dispatch.<br>• Refactored codebase, removed redundant app.py, built run_app.bat, configured .vscode. | **High (Lead)** |
| **Krishu Kumar**<br>Branch: krishu | • Authored initial Streamlit dashboard application (commits fca04dc, fd8d996).<br>• Integrated the Indian Roads Dataset (20,000 records).<br>• Authored Google Colab exploratory research notebook (98f279f).<br>• Configured Linux system packages (packages.txt) and unpinned libraries. | **High** |
| **Mayank Gupta**<br>Branch: mayank | • Added foundational IoT Simulation & Alert Monitoring module (commit 269d0c1).<br>• Created initial iot_simulator.py and pages/iot_monitor.py.<br>• Integrated Pydeck and Numpy requirements for geospatial plotting. | **High** |
| **Alok Chaubey**<br>Branch: alok | • Resolved syntax errors and deduplicated sensor generator functions (commit 5f14dde).<br>• Synchronized state variables between streamlit_app.py and multi-page components.<br>• Added Dev Container environment configuration (e5dfaf4). Merged PR #8. | **High** |
| **Arvind Mourya**<br>Branch: arvind | • Implemented State-Wise Accident Analysis module (commit 567d35e).<br>• Built grouping aggregations calculating accident volume and casualties for Top 10 Indian states.<br>• Integrated dual bar charts comparing state frequencies and casualty ratios. | **High** |
| **Amit Kohli**<br>Commit: c8f4245 | • Engineered Accident Risk & Safety Insights dashboard banner (commit c8f4245).<br>• Added threshold-based visual alerts: High Risk (≥0.7), Moderate (≥0.4), and Lower Risk.<br>• Implemented adverse weather keyword counters for road safety advisories. | **High** |
| **Ayush Patel**<br>Branch: ayush | • Managed Pull Request #4 integration into repository (commit c108d2c).<br>• Synchronized updates to streamlit_app.py (commit 396683d).<br>• Contributed project setup instructions and feature descriptions in README.md (2c3227a). | **High** |
| **Pritam Kumar**<br>Branch: pritam | • Enhanced iot_simulator.py sensor generator methods (commit 0aa5a8d).<br>• Refined pages/iot_monitor.py alert display cards and UI styling (commit 03be2bc).<br>• Supported multi-page navigation routing and session persistence. | **Medium** |
| **Kritika**<br>Branch: kritikaaa | • Authored CONTRIBUTING.md team setup guide (commit 17c34b4).<br>• Verified IoT simulation telemetry parameters and supported documentation. | **Medium** |
| **Sunal**<br>Branch: sunal | • Updated iot_monitor.py parameters (commit b54a36b).<br>• Authored multiple README.md documentation iterations (f81277d, 82a8fdd, fff9bae). | **Medium** |
| **Daniyal Malik**<br>Branch: Daniyal | • Standardized notebook filename to Road_Accident_Pattern_Analysis.ipynb (commit 3758353).<br>• Conducted dataset schema verification. | **Medium** |
| **Riddhi Gupta**<br>Branch: Riddhi-Gupta | • Documented project feature overviews in README.md (commits 91db893, bad32ef).<br>• Researched empirical accident patterns and environmental hazard factors in Indian cities. | **Medium** |
| **Ayan Ansari**<br>Branch: ayan | • Managed PR #3 merge from Krishu (commit 434d47e).<br>• Contributed to deployment documentation updates in README.md (d6668d4). | **Low / Partial** |
| **Om**<br>Branch: om | • Participated in peer code review cycles, tested sidebar filter widgets, and assisted in UI responsiveness verification. | **Low / Minimal** |

# 8. Conclusion & Future Roadmap

**Conclusion:** The project successfully establishes an industrial-grade safety architecture unifying historical empirical pattern recognition with real-time IIoT physical actuation. By enforcing a clean single-entrypoint deployment on Streamlit Community Cloud, the system provides municipal authorities and traffic police with an accessible, high-performance platform for active accident mitigation.

**Future Engineering Scope:**

- **Hardware Edge Nodes:** Flashing physical ESP32/Raspberry Pi microcontrollers to transmit real-world LoRaWAN telemetry into the edge gateway broker.
- **Computer Vision Integration:** Deploying YOLOv8 on roadside CCTV video feeds for automated vehicle collision detection.
- **Vehicle-to-Everything (V2X):** Broadcasting active hazard advisories directly into vehicular telematics dashboards.